Ch 05: Value Returning Functions, Python Modules

Application Program Development

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Value-Returning Functions

Value-Returning Function

A value-returning function is a function that returns a value back to the caller. Python, as well as most other programming languages, provide a library of prewritten functions that perform commonly needed tasks.

```
def value_returning_function(parameter1, parameter2, ...):
    calculation1
    calculation2
    return result

r = value_returning_function(p1, p2, ...)
```

- **void function**: group of statements within a program for performing a specific task
  - Call function when you need to perform the task
  - The task is done as a *side-effect* of calling a void function
- **Value-returning function**: In contrast a value-returning function calculates a result from input and returns the result from calling the function
  - Value returned to the caller that invoked the function



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Standard Library Functions and the `import` Statement

- **Standard library**: library of pre-written functions that comes with a programming language like Python
  - **Library functions** perform tasks that programmers commonly need
    - Example: `print()`, `input()`, `range()`
    - Viewed by programmers as a "black box"
- Some library functions built into Python interpreter
  - To use, just call the functions

- **Modules**: collections of functions in standard library
  - Help organize library functions not built into the interpreter
  - Module **namespaces**
- To call a function stored in a module, need to write an `import` statement
  - Written at the **top** of the program
  - Format: `import module_name`



Figure 1: A library function viewed as a black box (Gaddis, 2021, pg.273)

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Generating Random Numbers (1 of 2)

- Random numbers are useful in a lot of programming tasks
- **random** module: includes library functions for working with random numbers
- **Dot notation** notation for calling a function in a module namespace
  - Format: `module_name.function_name()`
- `randint()` **function**: generates a random number in the range provide by the arguments
  - Returns the random number ot part of program that called the function
  - Returned integer can be used anywhere that an integer can
  - You can experiment with the function in interactive mode

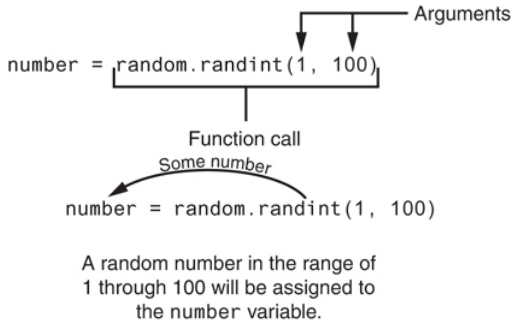


Figure 2: The random function returns a random int in asked for range (Gaddis, 2021, pg.274)

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Generating Random Numbers (2 of 2)

- **`randrange()` function**: similar to `range()` function, but returns randomly selected integer from the resulting sequence
  - Same arguments as for the `range()` function
- **`random()` function**: returns a random `float` in the range of 0.0 to 1.0
  - Does not have any arguments
- **`uniform()` function**: returns a random `float` but allows user to specify range

Pseudo-Random Numbers

Random number created by functions in `random` module are actually pseudo-random numbers.

- **Seed value**: initializes the formula that generates random numbers
  - Need to use different seeds in order to get different series of random numbers
    - By default uses system time for seed
    - Can use `random.seed()` function to specify desired seed value



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Writing Your Own Value-Returning Functions

Value-Returning Function

A value-returning function has a `return` statement that returns a value back to the caller of the function.

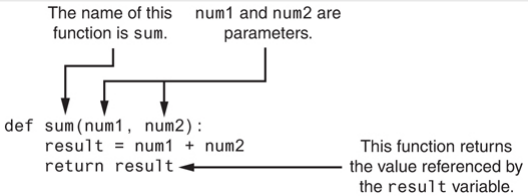


Figure 3: Parts of a function (Gaddis, 2021, pg.283)

- To write a value-returning function, you write a simple function and add one or more `return` statements
  - Format: `return expression`
    - The value for `expression` will be returned to the caller of your function
  - The expression in the `return` statement can be a complex expression, such as a sum of two variables or the result of another value-returning function

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Advantages of Value-Returning Functions

- Value-returning functions have the same benefits of functions we have discussed
  - simplify code
  - reduce duplication
  - provide for code reuse
- A major advantage of functions is increased ability to test code
  - A value-returning function is a black box: "data in, result out"
  - Can create tests of usual and edge cases for functions, to determine they work correctly and as expected
  - Tests are valuable products of software engineering systems, they give confidence code is working, and still working as it is modified

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Using IPO Charts

- **IPO chart**: describes the input, processing and output of a function
  - Tool for designing and documenting functions
  - Typically laid out in columns
  - Usually provide brief descriptions of input, processing, and output, without going into details
    - Often includes enough information to be used instead of flowchart

The `get_regular_price` Function

Input	Processing	Output
None	Prompts the user to enter an item's regular price	The item's regular price

The `discount` Function

Input	Processing	Output
An item's regular price	Calculates an item's discount by multiplying the regular price by the global constant `DISCOUNT_PERCENTAGE`	The item's discount

Figure 4: IPO charts for the get_regular_price() and discount() functions (Gaddis, 2021, pg.283)



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Tests and Python Docstring documentation

```python
import math
import pytest

# The sum function accepts two numeric arguments and
# returns the sum of those arguments.
def area_of_circle(radius):
    """
    Given the radius of some circle, calculate and
    return the area of the circle.

    Parameters
    ----------
    radius : float
        The radius of a circle we are being asked
        to calculate the area for.

    Returns
    -------
    area : float
        The area of the indicated circle is
        calculated and returned as the result of
        this function.
    """
    area = math.pi * radius**2.0
    return area
```

- Python Docstrings, many styles same as IPO but self-documenting code Python Docstrings (GeeksforGeeks, 2025)
- pytest Tutorial: Effective Python Testing (Hillard, 2024)

```python
# example of pytest tests
def test_zero():
    area = area_of_circle(0.0)
    assert area == pytest.approx(0.0)

def test_unit_circle():
    area = area_of_circle(1.0)
    assert area == pytest.approx(math.pi)

def test_small_circle():
    area = area_of_circle(0.5)
    assert area == pytest.approx(0.7853981633974483)
```

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Returning Boolean Values

- **Boolean function**: returns either `True` or `False`
  - Use to test a condition such as for decision and repetition structures
    - Common calculations, such as whether a number is even, can be easily repeated by calling a function
  - Use to simplify complex input validation code

```
def is_invalid(model_num):
    if model_num != 100 and model_num != 200 and model_num != 300:
        status = True
    else:
        status = False
    return status

# Get the model number.
model = int(input('Enter the model number: '))
# Validate the model number.
while is_invalid(model):
    print('The valid model numbers are 100, 200 and 300.')
    model = int(input('Enter a valid model number: '))
```

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Returning Multiple Values

- In Python, a function can return multiple values
  - Specified after the `return` statement separated by commas
  - Format: `return expression1, expression2, etc.`
- When you call such a function in an assignment statement, you need a separate variable on the left side of the = operator to receive each returned value

```
def get_name():
    # Get the user's first and last names.
    first = input('Enter your first name: ')
    last = input('Enter your last name: ')
    # Return both names.
    return first, last

first_name, last_name = get_name()
```



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The `math` Module

- **`math` module**: part of standard library that contains functions that are useful for performing mathematical calculations
  - Typically accept one or more values as arguments, perform mathematical operation, and return the result
  - Use of module requires an `import math` statement
- The `math` module defines variables `pi` and `e` which are assigned the mathematical values for π and e
  - Can be used in equations that require these values, to get more accurate results
- Variables must also be referred to using the dot notation
  - Example: `circle_area = math.pi * radius**2.0`

Python `math` Module

The Python standard library's `math` module contains numerous functions that can be used in mathematical calculations.



The math Module

Table 1: Many of the functions in the math module

math Module Function	Description
acos(x)	Returns the arc cosine of x in radians
asin(x)	Returns the arc sine of x in radians
atan(x)	Returns the arc tangent of x in radians
ceil(x)	Returns the smallest integer that is greater than or equal to x
cos(x)	Returns the cosine of x in radians
degrees(x)	Assuming x is an angle in radians, the function returns the angle converted to degrees
exp(x)	Returns e^x
floor(x)	Returns the largest integer that is less than or equal to x
hypot(x, y)	Returns the length of a hypotenuse that extends from (0, 0) to (x, y)
log(x)	Returns the natural logarithm of x
log10(x)	Returns the base-10 logarithm of x
radians(x)	Assuming x is an angle in degrees, the function returns the angle converted to radians
sin(x)	Returns the sine of x in radians
sqrt(x)	Returns the square root of x
tan(x)	Returns the tangent of x in radians



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Storing Functions in Modules

Python Modules

A module is a file that contains Python code. Large programs are easier to debug and maintain when they are divided into modules.

A module defines a **namespace** within which all functions and variables declared in the module reside and are accessed from.

- In large, complex programs, it is important to keep code organized
- **Modularization**: grouping related functions in modules
  - Makes program easier to understand, test and maintain
  - Make it easier to reuse code from multiple different programs
    - Import the module containing the required function to each program that needs it
- Modules contain function definitions but does not contain calls to the functions
  - Importing programs will call and use the functions in a module
- Rules for module names:
  - File name should end in `.py`
  - Cannot be the same as a Python keyword
- Import module using `import` statement

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Summary

- **Value-returning functions** take input, perform a calculation, and return the result of the calculation
- Python contains a large and useful **standard library** that can be `imported` and used in your programs
  - When you `import` a module, it will be in a **namespace** defined by the module name
- You can write your own value-returning functions using the `return` keyword to return a value or expression
- Advantages of functions:
  - simplify code
  - reduce duplication
  - provide for code reuse
  - increase ability to test code
- **Modularization** is the grouping of related functions in modules, the standard library is a collection of useful modules that you can use



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References

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- Hillard, D. (2024). *Pytest tutorial: Effective python testing*. Retrieved December 8, 2024, from https://realpython.com/pytest-python-testing/

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